MoS_2-Enhanced Antisymmetric 2T2R Logic Unit for One-Step Write/Compute In-Memory XNOR in BNN Accelerators

Zhoujie Pan, Yuwan Hong, Dingyi Zhang, Yanming Liu, and He Tian

***Abstract*—The rise of artificial intelligence (AI) and growing energy constraints have intensified the need for efficient, low-power hardware. The XNOR logic, a cornerstone of low-precision neural network inference, is especially critical to optimize for energy efficiency. Conventional CMOS-based in-memory XNOR units, however, face key limitations, including low integration density, readout interference, and multistep write protocols. To address these challenges, we present a MoS_2 transistor-enhanced antisymmetric 2T2R in-memory XNOR logic unit, enabling both write and compute operations to be finished within a single step, achieving the theoretical minimum for in-memory logic primitives. Beyond logic-level optimization, the incorporation of MoS_2 transistors facilitates monolithic 3-D integration and lowers static power consumption relative to standard CMOS devices. At the system level, we extend the analog majority architecture to BNN inference, including the weight mapping strategy, and discuss the system-level advantages and quantization errors introduced by this approach.**

***Index Terms*—BNN, MoS_2, resistive random access memory (RRAM), XNOR logic.**

I. Introduction

The rapid advancement of artificial intelligence has driven unprecedented demand for computational power and energy efficiency [1], [2], [3], posing significant challenges for hardware-constrained deployments [4], [5]. In response, network compression techniques have emerged as a compelling approach, reducing the overhead of weight deployment and inference by quantizing both weights and activations to low-bit representations [6]. Among these, binarized neural networks (BNNs), where both weights and activations are binarized to a single bit, enable the replacement of multiply-accumulate (MAC) operations with highly efficient in-memory XNOR bitwise logic, offering substantial improvements in speed, energy efficiency, and memory utilization [1], [7].

As a fundamental computational unit for BNN inference, optimizing the efficiency of in-memory XNOR logic can directly lead to performance improvement at the system level. Recent research has focused on advancing in-memory XNOR logic through diverse device and architectural innovations, aiming to maximize its write/compute efficiency and scalability for next-generation edge artificial intelligence (AI) systems [8], [9], [10].

However, current in-memory XNOR logic units based on conventional CMOS technology face significant limitations. For instance, spin-transfer torque-magnetic random access memory (STT-MRAM) alternatives are plagued by low ON/OFF ratios and analog MAC compatibility as array scale expands [8], while SRAM- and DRAM-based implementations suffer from volatility, integration density, and potential static power overhead [9], [10]. Comparing with conventional architectures [11], [12], resistive random access memory (RRAM) features excellent compatibility with high-efficiency analog compute-in-memory (CIM) architectures and has attracted growing attention [13].

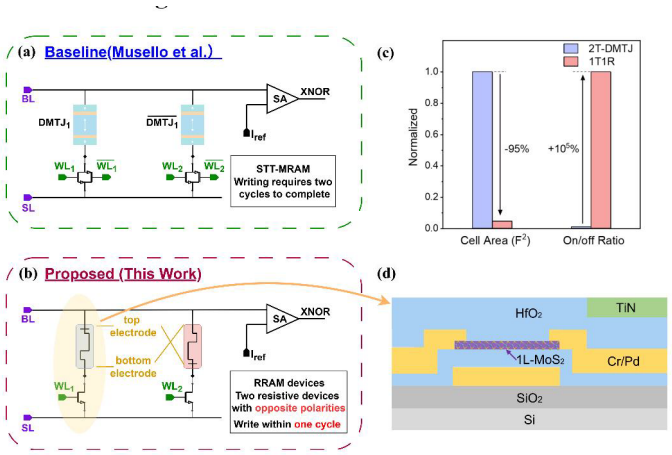


Fig. 1. Hardware implementation proposed in (a) [8] and (b) this work (optimized) using RRAM bitcells with opposite polarities for one-step write/compute XNOR operation. (c) Comparative summary of key metrics for the RRAM-based 1T1R cell and the 2T-DMTJ cell, when normalized to the same technology node. (d) Schematic of MoS_2-based 1T1R with positive polarity.

In this work, we report a one-step write/compute CIM XNOR operation units using MoS_2-based antisymmetric 2T2R scheme, as shown in Fig. 1(b), that addresses key limitations of conventional CMOS-based implementations. Leveraging the excellent compatibility of MoS_2 transistors with monolithic 3-D integration, our proposed design achieves higher integration density, an improved switching ratio, and a simplified fabrication process enabled by the introduction of RRAM. Moreover, the antisymmetric architecture allows both computation and write operations to be completed within a single cycle, which reaches the theoretical lower bound for in-memory XNOR logic. This capability supports ultrafast weight deployment, making it particularly suitable for edge AI applications. Furthermore, our proposed structure provides a solution for efficient in-memory XNOR operations at the system level, including applications in BNN accelerators and Ising solvers [14]. We extend the analog majority architecture used in [8] to the more complex BNN inference, including the weight mapping strategy, and discuss the system-level advantages and quantization errors introduced by this approach [15]. The difference between our work and baseline is shown in Fig. 1.

II. Our Proposed Optimized XNOR Bitcell

A. MoS_2 Transistor-Enhanced Antisymmetric 2T2R Bitcell

Exploiting the advantage of simple stacked structure of memristors, we are able to fabricate an antisymmetric 2T2R unit with a four-

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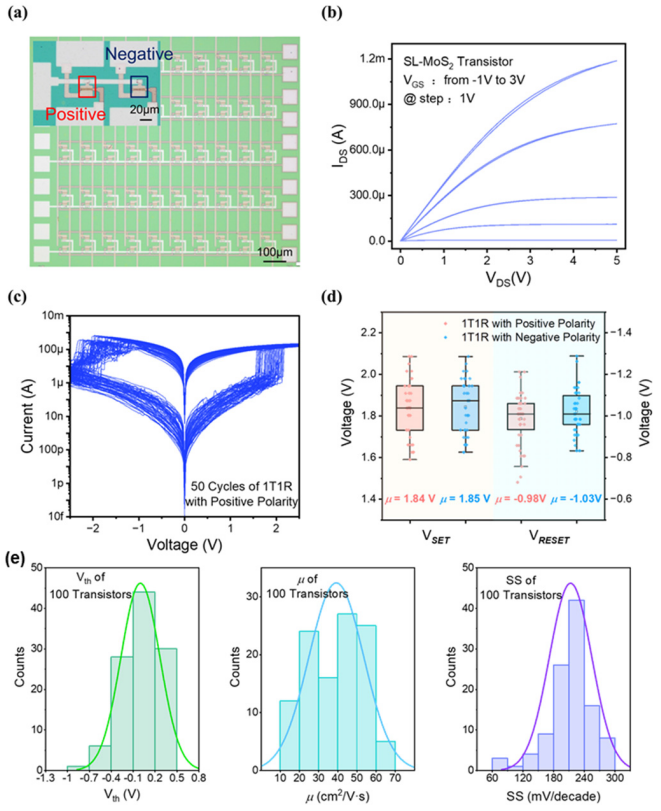


Fig. 2. (a) Optical microscope images of the MoS_2-based XNOR-bitcount and a 5 × 10 array. (b) Measured I_D–V_{DS} characteristics of a 2-D MoS_2 FET at different V_{GS}. (c) 50 DC sweep cycles of 1T1R with positive polarity, demonstrating stable memristive behavior. (d) Statistical distribution of set and reset voltages for 1T1R configurations with positive and negative polarity, respectively, showing high uniformity. (e) Transfer characteristics of MoS_2 FETs, including field-effect mobility, threshold voltage, and subthreshold swing (SS).

step process after the fabrication of transistors: first, the cathode of the positive-polarity RRAM—typically an inert metal—is shared with the transistor contact layer; next, the anode (TiN) for the negative-polarity RRAM is deposited; then, a common resistive switching layer is grown for both devices; finally, the TiN electrode for the positive device and the inert electrode for the negative device are separately formed. In order to experimentally validate the proposed scheme, we fabricated a batch of devices for statistical characterization. The optical microscope images of a single MoS_2-based XNOR 5 × 10 2T2R array are shown in Fig. 2(a). Fig. 2(b) shows the I_D–V_{DS} characteristics of a 2-D MoS_2 FET at different V_{GS}, demonstrating considerable driving current and a high ON/OFF ratio. Fifty consecutive DC cycles of the 1T1R device demonstrate excellent cycle-to-cycle stability [Fig. 2(c)], from which the set and reset threshold voltages were extracted. Fig. 2(d) presents the statistical distributions of these voltages for 40 devices under both positive and negative polarities. The robust polarity insensitivity greatly simplifies the control of the 2T2R configuration. The key electrical parameters of MoS_2 FETs are summarized in Fig. 2(e).

B. One-Step Write/ Compute XNOR Operation

According to the architecture proposed in [8], the writing process of each bitcell storing a complementary weights pair requires a two-step operation due to the unavoidable reset step.

Therefore, a single XNOR operation for one bitcell requires two cycles to complete the writing process. The manual reset operation not only limits the writing speed but also increases energy consumption.

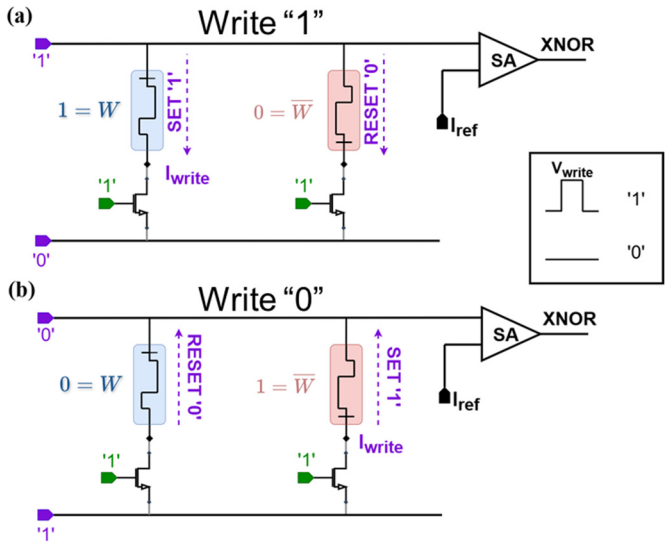


Fig. 3. (a) and (b) Explain the operation method for writing W = "1" and W = "0" in our proposed XNOR-BC units.

1) Our methodology of XNOR logic computation still aligns with the traditional method, XNOR = $A \cdot W + \bar{A} \cdot \bar{W}$. However, their weight writing process, Step-1 and Step-2 in the weight-writing process, can be merged into one step adopting our proposed architecture. Optimized steps of one XNOR structure are as follows.

1) *Write Weights:* In one cycle, write expected weights, W, and $\bar{W}$, into two devices simultaneously by controlling the input of two WLs (when writing W, $\bar{W}$ will be reset because of the opposite polarities design; inversely for writing $\bar{W}$). Detailed information is discussed in Fig. 3.

2) *AND + OR Operation*: Apply input activations A and $\bar{A}$ to the WLs of two RRAM devices separately and obtain the XNOR result current from the BL. A more detailed case-by-case discussion is provided in Fig. 4.

To execute the multibit XNOR-BC operation, we can adopt the same parallel XNOR computations and "analog majority" approach. By comparing the accumulated current $I_{N-XNORs}$ generated by N XNOR-BCs with I_{N-ref} in the SA, the overall computation result can be secured.

C. Performance Demonstration and Comparison

Combining the statistical trends extracted from device-level experimental results with the previously proposed logic scheme, we further explore the design space via SPICE simulations to identify bias conditions that simultaneously ensure logic robustness and reduce read/write energy consumption. The corresponding SPICE waveforms are presented in Fig. 5.

Compared with other XNOR operations implemented in computing-in-memory architecture reported in previous studies (see [8], [18], [19], [20], [21], [22], [23]), this work is the first to finish the write operation and computation operation both in a single step, as shown in Fig. 6(a). Beyond circuit-level optimization, the MoS_2-based 2T2R with opposite polarities leverages the inherent compatibility of 2-D materials with monolithic 3-D integration, enabling the realization of a compact 3-D architecture as schematically illustrated in Fig. 6(b). In a planar configuration, the fabrication requires two additional material deposition steps, and as for the traditional CMOS approach, the multistacking is limited to the package level. On the contrary, the MoS_2-based stacked 2T2R architecture allows the integration of both forward- and reverse-polarity memristors within a unified process flow. This integration not only reduces the number of process steps

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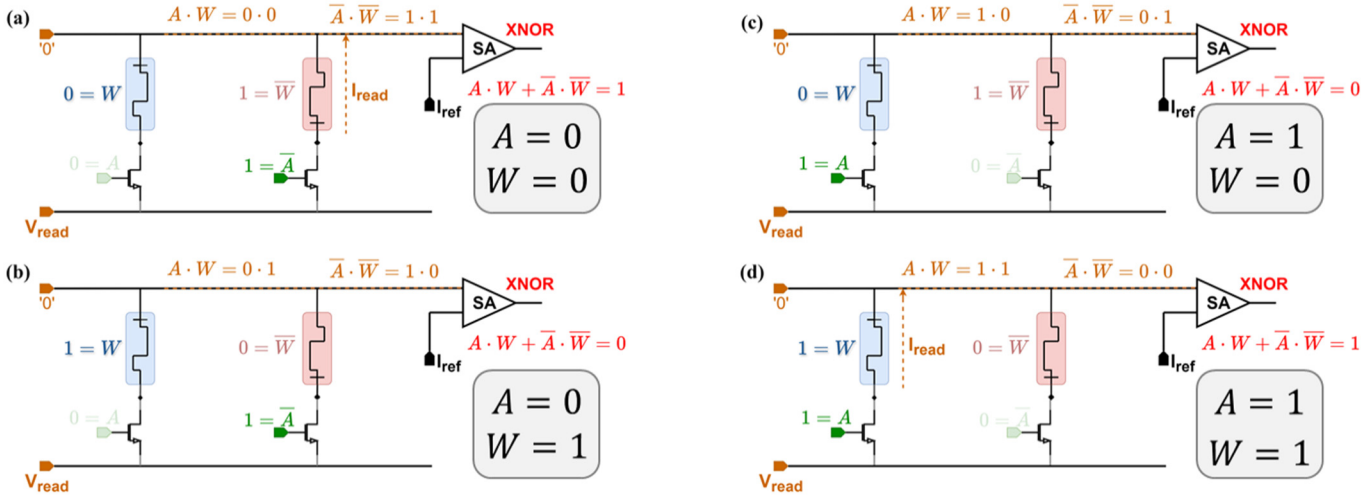


Fig. 4. (a)–(d) Detailed processes of a total of four possible XNOR operations corresponding to different weights and input activation values using the proposed self-resetting architecture.

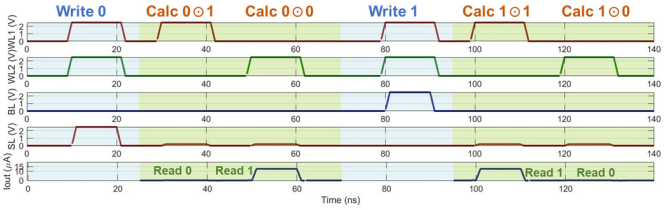


Fig. 5. Bias voltages applied to complete the WRITE and XNOR operations presented in Figs. 3 and 4. The waveforms corresponding to the write and compute operations of our XNOR unit are also illustrated.

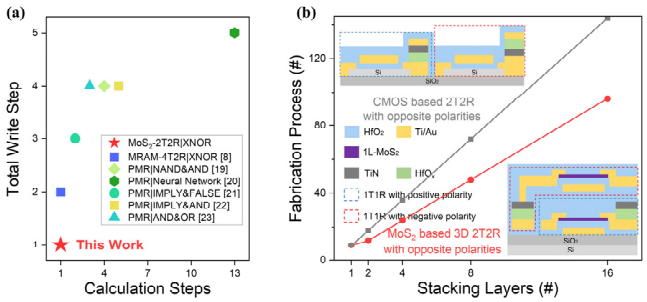


Fig. 6. (a) Comparison of the performance of this work with other XNOR-BC devices in terms of "computation steps" and "total writing steps." (b) Comparison of multilayer stacking process steps between MoS_2-based 2T2R and CMOS-based 2T2R.

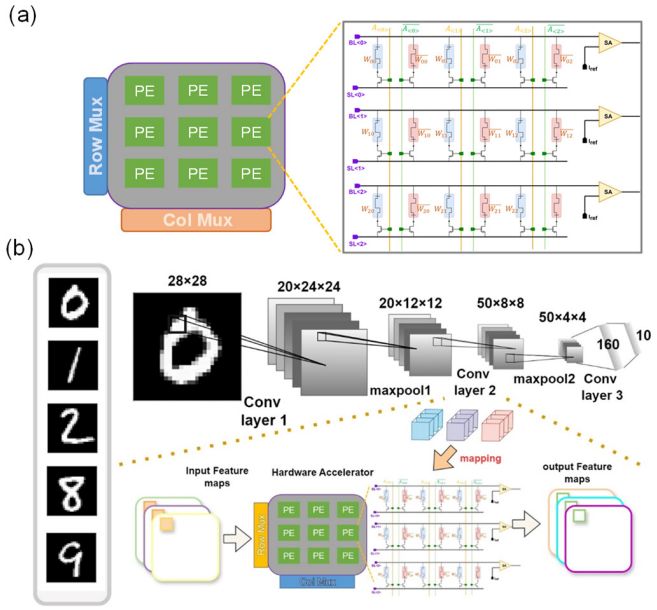


Fig. 7. (a) Convolutional kernel hardware implementation based on our proposed XNOR logic cell. (b) Simulated binary neural network inference workflow using LeNet, including weight quantization, deployment, and hardware-aware inference simulation.

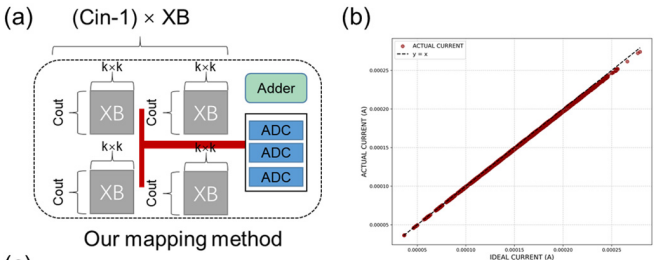
but also compacts the footprint to half of the original area [Fig. 6(b)]. Although our MoS_2-based 2T2R cell remains constrained by reading delay and energy, 3-D integration can be employed to enhance its computational parallelism, thereby improving the energy efficiency of computation.

III. System-Level Application: Toward Binary Neural Network Acceleration

The MAC operation is a fundamental component of any neural network: it computes the weighted sum of inputs, followed by an accumulation of the results. In CNNs, each filter (a $k \times k$ weight matrix) is convolved with each feature map (input activation matrix). However, the computational intensity of the MAC operation increases significantly with the size of the network and its parameters, which limits the efficiency of the model. To address this challenge, the BNN is introduced, where both the weights and activations are binarized to "−1" (representing logic "0") or "+ 1" (representing logic "1"). This binarization simplifies the MAC operation into a bitwise XNOR operation [17]: instead of performing multiplications, bitwise XNORs between weights and activations are computed, constraining the entire process and results with binary values of "0"s and "1"s. Through this simplification, the computational cost and memory requirements are significantly reduced. Similar to the approach proposed in [8], our method can also be readily extended to in-memory computation of activation values in BNNs.

In order to further demonstrate the application potential of our work, we further build a system-level XNOR-BNN hardware-aware simulator (code implementation details can be found in [24]). Fig. 7(a) and (b) demonstrates a hardware implementation method

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(c)

	Technology	Memory	Resolution	Cycle time	Energy efficiency
This work with analog majority	65nm	RRAM	In: 1 W:1 O:1	10ns	158.61 TOPS/W
This work with Full precision	65nm	RRAM	In: 1 W:1 O:5	10ns	52.98 TOPS/W
IEEE SSCL 24[25]	40nm	RRAM	In: 1 W:1 O:6	11.1ns	7.8~58.5 TOPS/W
JSSC 21[26]	40nm	RRAM	In: 1~8 W:1~8 O:20*	10ns	56.67 TOPS/W
ISSCC 21[27]	22nm	SRAM	In: 4 W:4 O:16	10ns	89 TOPS/W

Fig. 8. (a) Our mapping method, with XB array size of (k × k, Cout) (k: the kernel size of the convolution layer, and Cin and Cout: the number of in/out channels of the convolution layer. (b) Ideal current compared with actual current caused by array nonidealities. (c) Energy efficiency comparison.

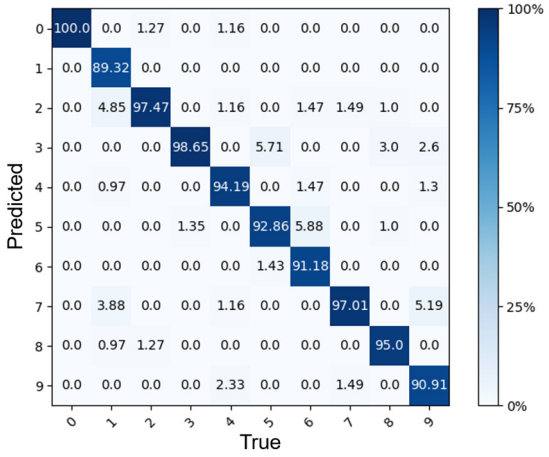


Fig. 9. Simulated accuracy result confusion matrix.

based on our design, by mapping weights into a crossbar, we are able to execute the XNOR operation in parallel in-memory. Leveraging the lightweight characteristics of binary neural networks, we quantize the computational bottleneck layers of the LeNet network, wherein the convolutional inference can be substituted by the XNOR-bitcount scheme proposed in this work, thereby demonstrating the system-level application potential of our approach. Due to the fast write nature of our architecture, our hardware also has the potential to quickly update weights, making it suitable for performing edge tasks.

We also extend the analog majority scheme used in [8] to BNN inference scenarios. As shown in Fig. 8(a), our weight mapping scheme is an improvement upon the convolutional neural network RRAM acceleration architecture proposed in ISAAC [28]. We adopt an input-channel-splitting strategy for convolution operations to mitigate size-dependent nonidealities (such as IR drop) and enable the extension of analog majority. The impact of array nonidealities after the input-channel-splitting strategy is simulated by SPICE, and the result indicates an average relative error of 1.16%, which does not interfere with the judgment of the analog majority, as shown in Fig. 8(b). Fig. 8(c) shows the energy efficiency comparison table. Since the analog majority scheme reduces the precision requirements for ADCs, thereby optimizing the significant ADC energy consumption within the system, the proposed extended analog majority architecture achieves notable energy efficiency advantages while maintaining BNN inference accuracy. Meanwhile, we acknowledge that while the impact of device variability is controllable at the network scale adopted in the present work, relevant existing studies [29], [30] can be leveraged to mitigate such variability effects as future scenarios expand toward energy-efficient inference in larger scale networks.

Our results show that under the proposed weight mapping strategy, even when applying 1-bit quantization via the analog majority scheme for the output of each input channel, the network accuracy remains above 95%, as shown in Fig. 9. Moreover, our network is also validated on the more complex CIFAR-10 dataset and shows an accuracy of 86.40%.

IV. Conclusion

In this work, we present a one-step write/compute in-memory XNOR logic unit based on a MoS_2 transistor-enhanced antisymmetric 2T2R scheme. Our work reaches a theoretical minimum for in-memory logic, doubling the write efficiency compared to prior state-of-the-art approaches. We further propose a novel architecture based on our MoS_2-based 2T2R, leveraging the monolithic integrability of 2-D materials to achieve higher integration density.

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